HF Skywave Massive MIMO Communications with Interference Sparsity-Aware Turbo Receiver

Linfeng Song*†, Rui Sun*†, Ding Shi†, Yanbo Yu*, An-An Lu*†, Xiqi Gao*†, Geoffrey Ye Li‡, and Xiang-Gen Xia§

*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China

†Purple Mountain Laboratories, Nanjing 211111, China

‡Department of Electrical and Electronic Engineering, Imperial College London, London SW7 2AZ, U.K.

§Departement of Electrical and Computer Engineering, University of Delaware, Newark, DE 19716 USA

Email: songlf@seu.edu.cn, ruisun@seu.edu.cn, shiding@pmlabs.com.cn, 220230863@seu.edu.cn, aalu@seu.edu.cn, xqgao@seu.edu.cn, geoffrey.li@imperial.ac.uk, xxia@ee.udel.edu

2026 IEEE Wireless Communications and Networking Conference (WCNC) | 979-8-3315-7729-2/26/$31.00 ©2026 IEEE | DOI: 10.1109/WCNC65185.2026.11555216

***Abstract*—In this paper, we propose a low complexity turbo receiver for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) systems. By leveraging the beam based channel model (BBCM), we reveal the interference sparsity of HF skywave massive MIMO systems. Exploiting the interference sparsity, we provide a condition of extracting sufficient observation for signal detection. Motivated by this condition, we construct the interference user terminal (UT) set (IUS) and extract the observation vector from the received signal after matched filtering (MF) for each UT. Then, a low-dimensional interference sparsity-aware detector (ISD) is separately designed for each UT, and the interference sparsity-aware turbo receiver (ISTR) is subsequently formulated using ISDs. Further, we develop an efficient implementation of the ISTR, involving approximate computations of the ISD and the signal reconstructed by ISD. Simulation results confirm that the proposed ISTR achieves excellent performance with relatively low complexity.**

***Index Terms*—Massive MIMO, HF skywave communications, beam structured turbo receiver.**

I. Introduction

With the continuous increase in the demand for global coverage [1], high frequency (HF) skywave communications have surfaced as a technology of a great promise. Nevertheless, in traditional point-to-point HF communication scenarios, the data rates remain relatively low. To address this issue, introducing massive multiple-input multiple-output (MIMO) into HF skywave communications holds profound significance. The concept of HF skywave massive MIMO has been proposed in [2] and developed in [3], [4], etc., confirming the effectiveness of massive MIMO for HF skywave communications.

For massive MIMO transmission, receiver design is a crucial issue. Linear receivers are extensively applied in various communication scenarios, such as the matched filtering (MF), zero-forcing (ZF), and minimum mean-squared error (MMSE) receivers. However, the above linear receivers acheive low performance especially when with ill-conditioned channel matrix [5], and ZF or MMSE receiver involves high dimensional matrix inversions for massive MIMO. For performance improvement, the framework of turbo receiver can be adopted by exchanging soft information between a soft-input soft-output (SISO) detector and a SISO decoder [6]. For complexity reduction, some works utilize the beam domain channel sparsity in receiver designs [7], [8]. In the context of HF skywave massive MIMO, the beam domain sparsity also has been introduced for low-complexity receiver designs [9], [10]. In addition to the beam domain channel sparsity, the HF skywave massive MIMO also exhibits interference sparsity, which provides a new perspective for low-complexity receiver design.

In this paper, we investigate the turbo receiver design for HF skywave massive MIMO exploiting interference sparsity. Firstly, we reveal the interference sparsity for HF skywave massive MIMO systems using the beam based channel model (BBCM). Secondly, we construct the interference user terminal (UT) set (IUS) with interference sparsity and design a low-dimensional interference sparsity-aware detector (ISD) for each UT, then the interference sparsity-aware turbo receiver (ISTR) is developed by incorporating ISDs. Finally, we propose an efficient implementation of ISTR by using the structural properties of the beam matrix and the sparsity of the beam domain channel in BBCM.

II. System Model

A. System Configuration

Consider an HF skywave massive MIMO orthogonal frequency division multiplexing (OFDM) system with carrier frequency f_{c} and subcarrier spacing Δ_{f}. And U single-antenna UTs are served by the BS with M-antenna uniform linear array (ULA). In HF skywave communication systems, the carrier frequency, f_{c}, needs to be adjusted when the ionospheric environment changes [11] while the inter-antenna spacing cannot be configured accordingly. To this end, different from the half wavelength of carrier in traditional terrestrial cellular massive MIMO communications, the inter-antenna spacing of

This work was supported by the Jiangsu Province Major Science and Technology Project under Grant BG2024005, the Fundamental Research Funds for the Central Universities under Grant 2242022k60007, the SEU Innovation Capability Enhancement Plan for Doctoral Students under Grant CXJH_SEU 26091, the Basic Research Program of Jiangsu under Grant BK20250290, and the Jiangsu Funding Program for Excellent Postdoctoral Talent under Grant 2025ZB025.

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.

HF skywave massive MIMO is typically set as $d = \lambda_{\rm o}/2$, where $\lambda_{\rm o} = c/f_{\rm o}$ with the speed of light c, and $f_{\rm o} > f_{\rm c}$ is denoted as the highest system operating frequency.

B. Beam Based Channel Model

We assume the channel states stay constant in each OFDM symbol and focus on the uplink (UL) transmission over one OFDM symbol. The data transmitted by the u-th UT at the s-th subcarrier is denoted as $x_{u,s}$. Consequently, after OFDM demodulation, the signal received by the m-th antenna at the s-th subcarrier is given as

$$y_{m,s} = \sum_{u=1}^{U} h_{u,m,s} x_{u,s} + z_{m,s}, \tag{1}$$

where $z_{m,s} \sim \mathcal{CN}(0, \sigma_{\rm z})$ is the complex additive white Gaussian noise, $h_{u,m,s}$ represents the channel frequency response at the s-th subcarrier between the UT u and the m-th antenna at the BS. Stacking the channel frequency responses of all antennas yields the space domain channel vector $\mathbf{h}_{u,s} \triangleq [h_{u,1,s}, \cdots, h_{u,M,s}]^{\rm T} \in \mathbb{C}^M$, which can be given as [9]

$$\mathbf{h}_{u,s} = \sum_{p=1}^{P_u} \alpha_{u,p} \exp\{-\bar{\jmath} 2\pi s \Delta_{\rm f} \tau_{u,p}\} \mathbf{v}(\phi_{u,p}), \tag{2}$$

with the steering vector

$$\mathbf{v}(\phi) \triangleq \frac{\omega(\phi)}{\sqrt{M}} [\exp\{-\bar{\jmath} 2\pi \phi\}, \cdots, \exp\{-\bar{\jmath} 2\pi M \phi\}]^{\rm T} \in \mathbb{C}^M. \tag{3}$$

In (3), $\omega(\phi) = \exp\{\bar{\jmath}\pi(M+1)\phi\}$, the scaled directional cosine $\phi = f_{\rm c}\Delta_\tau \Omega \in [-f_{\rm c}\Delta_\tau, f_{\rm c}\Delta_\tau)$, the directional cosine $\Omega \in [-1, 1)$, and $\Delta_\tau \triangleq d/c = 1/(2f_{\rm o})$. In (2), P_u is the number of paths from UT u to the BS, $\alpha_{u,p}$, $\tau_{u,p}$ and $\phi_{u,p}$ represent the complex gain, delay and scaled directional cosine of path p from UT u, respectively.

To achieve a sparse representation of the space domain channel, we perform uniform sampling for scaled directional cosine and introduce the beam domain channel since the angle spread of the HF skywave communication is particularly small. Specifically, we set the interval length for uniform sampling as $\Delta_\phi = 1/S$ with $S = FM$, where F is the fine factor, and symmetric sampling is considered. Consequently, the a-th sampled scaled directional cosine is obtained as $\phi_a = (a - 1 - \lfloor f_{\rm c}\Delta_\tau/\Delta_\phi \rfloor)\Delta_\phi$, and the range of scaled directional cosine $[-f_{\rm c}\Delta_\tau, f_{\rm c}\Delta_\tau)$ can be partitioned into $A = 2\lfloor f_{\rm c}\Delta_\tau/\Delta_\phi \rfloor + 1$ mutually exclusive subsets $\mathcal{S}_1, \cdots, \mathcal{S}_A$ with the a-th subset being the interval

$$\mathcal{S}_a \triangleq [\max\{-f_{\rm c}\Delta_\tau, \phi_a - \Delta_\phi/2\}, \min\{f_{\rm c}\Delta_\tau, \phi_a + \Delta_\phi/2\}). \tag{4}$$

By approximating $\mathbf{v}(\phi_{u,p})$ as $\mathbf{v}(\phi_a)$, if $\phi_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_a$, with $\mathcal{P}_u \triangleq \{\phi_{u,1}, \cdots, \phi_{u,P_u}\}$, the space domain channel in (2) can be modelled as

$$\mathbf{h}_{u,s} = \sum_{a=1}^{A} \underbrace{\sum_{\phi_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_a} \alpha_{u,p} \exp\{-\bar{\jmath} 2\pi s \Delta_{\rm f} \tau_{u,p}\}}_{= d_{u,a,s}} \mathbf{v}(\phi_a) = \mathbf{V} \mathbf{d}_{u,s}, \tag{5}$$

where $\mathbf{d}_{u,s} \triangleq [d_{u,1,s}, \cdots, d_{u,A,s}]^{\rm T} \in \mathbb{C}^A$ is the beam domain channel vector, and $\mathbf{V} \triangleq [\mathbf{v}(\phi_1), \cdots, \mathbf{v}(\phi_A)] \in \mathbb{C}^{M \times A}$ is termed as the beam matrix. Combining (3), the beam matrix can be expressed as

$$\mathbf{V} = \frac{1}{\sqrt{M}} \operatorname{diag}\{\boldsymbol{\lambda}_1\} \mathbf{I}_{M,S} \mathbf{F}_S \mathbf{I}_{S,A} \operatorname{diag}\{\boldsymbol{\lambda}_2\}, \tag{6}$$

where $\mathbf{F}_S = [\mathbf{f}_{S,1}, \cdots, \mathbf{f}_{S,S}] \in \mathbb{C}^{S \times S}$ denotes S-dimensional DFT matrix with $[\mathbf{F}_S]_{i,j} = \exp\{-\bar{\jmath} 2\pi (i-1)(j-1)/S\}$, $\mathbf{I}_{N_1,N_2}$ is constructed by the first N_1 columns ($N_1 \le N_2$) of $\mathbf{I}_{N_2}$ or the first N_2 rows ($N_1 > N_2$) of $\mathbf{I}_{N_1}$, and the phase-shift vectors $\boldsymbol{\lambda}_1 = \mathbf{I}_{M,S} \mathbf{f}_{S,S-(A-1)/2+1}$, $\boldsymbol{\lambda}_2 = \mathbf{I}_{A,2S} \boldsymbol{\Pi}_{2S}^{(A-1)/2} \mathbf{f}_{2S,M}^*$. According to (6), beam matrix $\mathbf{V}$ exhibits the structure of a phase-shifted partial DFT matrix. We call (5) as the beam based channel model (BBCM). It can be written in a more compact form

$$\mathbf{H}_s = \mathbf{V} \mathbf{D}_s, \tag{7}$$

where at the s-th subcarrier, the space domain channel matrix and the beam domain channel matrix are defined as $\mathbf{H}_s \triangleq [\mathbf{h}_{1,s}, \cdots, \mathbf{h}_{U,s}] \in \mathbb{C}^{M \times U}$ and $\mathbf{D}_s \triangleq [\mathbf{d}_{1,s}, \cdots, \mathbf{d}_{U,s}] \in \mathbb{C}^{A \times U}$, respectively.

Due to the narrow angle spread of HF skywave channels [2], the beam domain channel exhibits remarkable sparsity, that is, the number of non-zero elements of $\mathbf{d}_{u,s}$ is small. Denote the non-zero beam set $\mathcal{A}_u$ as the index set of non-zero elements in $\mathbf{d}_{u,s}$ with $A_u = |\mathcal{A}_u|$, then $A_u \ll A$. In this paper, we assume that beam domain channel $\mathbf{D}_s$ is obtained at the BS by channel estimation [3].

III. Interference Sparsity-Aware Turbo Receiver

A. Principal of Turbo Receiver

In the framework of turbo receiver, a bit-interleaved coded modulation (BICM) system is usually considered. At the transmitter, the original information bitstream is sequentially encoded by an encoder and interleaved by an interleaver. After interleaving, the bitstream is partitioned into vectors of length N and mapped to symbols in constellation $\mathbb{S} = \{s_1, \cdots, s_K\}$ with modulation order $K = 2^N$, where $s_k \in \mathbb{C}$ is the k-th constellation point mapped by the bit vector $\mathbf{b}_k = [b_{k,1}, \cdots, b_{k,N}]^{\rm T} \in \mathbb{B}^N$. For UT u at the subcarrier s, a bit vector after interleaving is denoted as $\mathbf{q}_u = [q_{u,1}, \cdots, q_{u,N}]^{\rm T} \in \mathbb{B}^N$ and mapped to the transmitted symbol $x_u \in \mathbb{S}$, where index s of subcarrier is omitted for brevity. Consequently, transmitted signal vector $\mathbf{x}$ for all UTs is given by $\mathbf{x} = [x_1, \cdots, x_U]^{\rm T} \in \mathbb{S}^U$, and the received signal in (1) can be written in a compact form

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}, \tag{8}$$

where received signal $\mathbf{y} \triangleq [y_1, \cdots, y_M]^{\rm T} \in \mathbb{C}^M$ and noise $\mathbf{z} \triangleq [z_1, \cdots, z_M]^{\rm T} \in \mathbb{C}^M$. It is assumed that $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\rm z}\mathbf{I})$. The a priori mean and variance of transmitted signal vector $\mathbf{x}$ are respectively denoted as $\boldsymbol{\mu} \triangleq \mathbb{E}\{\mathbf{x}\} \in \mathbb{C}^U$ and $\boldsymbol{\Sigma} \triangleq \operatorname{Cov}\{\mathbf{x}, \mathbf{x}\} \in \mathbb{R}^{U \times U}$, where $\boldsymbol{\Sigma}$ is a diagonal matrix, indicating the uncorrelatedness among the symbols transmitted

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.

by different UTs. At the BS, a turbo receiver is implemented to iteratively decode the information bits from all UTs. During the iteration, soft information of each bit is exchanged between a SISO detector and a SISO decoder. The soft information output by SISO detector is the extrinsic LLR of each bit, which is computed by the a priori LLR of each bit and the detection of each transmitted symbol.

Typically, for SISO MMSE detection, conditioned on observation vector $\mathbf{y}$, extrinsic LLR of $q_{u,i}$ can be obtained as [12]

$$L_{\mathrm{e}}(q_{u,i}|\mathbf{y}) = \ln\frac{\sum_{s_k\in\mathbb{S}_i^1}\exp\left\{-|\mu_u^{\mathrm{e}}-s_k|^2/\sigma_u^{\mathrm{e}}+\sum_{j\neq i}^{N}(b_{k,j}-1/2)L'_{\mathrm{a}}(q_{u,j})\right\}}{\sum_{s_k\in\mathbb{S}_i^0}\exp\left\{-|\mu_u^{\mathrm{e}}-s_k|^2/\sigma_u^{\mathrm{e}}+\sum_{j\neq i}^{N}(b_{k,j}-1/2)L'_{\mathrm{a}}(q_{u,j})\right\}}, \tag{9}$$

where $\mathbb{S}_i^b \triangleq \{s_k \in \mathbb{S}|b_{k,i} = b\},\ b \in \mathbb{B}, i \in \mathcal{Z}_N^+$, is the symbol set with the i-th mapping bit $b_{k,i}$ of s_k being b, and $\mathcal{Z}_N^+ \triangleq \{1,\cdots,N\}$. In (9), the a priori LLR $L'_{\mathrm{a}}(q_{u,i})$ is initialized by 0 for the first iteration, and updated via the soft information output by the SISO decoder for the subsequent iterations, and the extrinsic mean and the extrinsic variance of x_u are respectively given by $\mu_u^{\mathrm{e}} = (\widetilde{x}_u/\widetilde{\epsilon}_u - \mu_u/\sigma_u)\sigma_u^{\mathrm{e}}$ and $\sigma_u^{\mathrm{e}} = (1/\widetilde{\epsilon}_u - 1/\sigma_u)^{-1}$, where $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ are the signal reconstructed by the detector and the detection MSE, respectively, which are given by [12]

$$\widetilde{x}_u = \mathbf{r}_u^{\mathrm{H}}(\mathbf{y} - \mathbf{H}\boldsymbol{\mu}) + \mu_u, \tag{10a}$$

$$\widetilde{\epsilon}_u = \left(1 - \mathbf{r}_u^{\mathrm{H}}\mathbf{h}_u\right)\sigma_u. \tag{10b}$$

In (10), MMSE detector $\mathbf{r}_u = (\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^{\mathrm{H}} + \sigma_{\mathrm{z}}\mathbf{I})^{-1}\mathbf{h}_u\sigma_u$, $\mu_u = [\boldsymbol{\mu}]_u$, and $\sigma_u = [\boldsymbol{\Sigma}]_{u,u}$.

Subsequently, the a priori soft information $L_{\mathrm{a}}(c_{u,i})$ for SISO decoder is obtained by mapping $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ to $L_{\mathrm{a}}(c_{u,i})$ via deinterleaving $c_{u,i} = \Pi^{-1}(q_{u,i})$ with interleaver $\Pi(\cdot)$. Then, $L'_{\mathrm{e}}(c_{u,i})$ is obtained after SISO decoding, and the a priori soft information $L'_{\mathrm{a}}(q_{u,i})$ for the input of SISO detector is acquired by mapping $L'_{\mathrm{e}}(c_{u,i})$ to $L'_{\mathrm{a}}(q_{u,i})$ via interleaving $q_{u,i} = \Pi(c_{u,i})$. As a result, the probability of each information bit is calculated by

$$P(q_{u,i} = b) = \frac{1}{2}\left(1 + (2b-1)\tanh\left(L'_{\mathrm{a}}(q_{u,i})/2\right)\right). \tag{11}$$

Thus, the a priori mean and variance of the transmitted symbol x_u are respectively updated as

$$\mu_u = \sum_{k=1}^{K} s_k P(x_u = s_k), \tag{12a}$$

$$\sigma_u = \sum_{k=1}^{K} |s_k|^2 P(x_u = s_k) - |\mu_u|^2, \tag{12b}$$

where $P(x_u = s_k) = P(\mathbf{q}_u = \mathbf{b}_k) = \prod_{i=1}^{N} P(q_{u,i} = b_{k,i})$. To improve the BER performance, the updated a priori information is used for the next iteration of signal detection, i.e., the calculations of $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ for each UT u.

B. Interference Sparsity-Aware Turbo Receiver

To develope a low complexity turbo receiver, we focus on the process of signal detection. The goal of signal detection is to recover $\mathbf{x}$ based on observation $\mathbf{y}$, channel $\mathbf{H}$ and a priori statistics $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. When the average power of each transmitted symbol is normalized to unit, the values of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are initialized as $\mathbf{0}$ and $\mathbf{I}_U$, respectively, and are updated by (12) for SISO signal detection.

However, for massive MIMO communications, when the number of antennas at the BS is large, the dimensionality of $\mathbf{y}$ becomes high for directly performing signal detection, which brings unbearable complexity. Therefore, we try to find a low-dimensional observation vector for signal detection of each UT. Before proceeding, we adopt the following definition.

Definition 1 *Vector $\boldsymbol{\theta}_u$ is defined as a sufficient observation for the detection of x_u if*

$$I(x_u, \boldsymbol{\theta}_u) = I(x_u, \mathbf{y}), \tag{13}$$

where $I(\cdot,\cdot)$ is mutual information between random variables.

From Definition 1, for the detection of x_u, observation $\boldsymbol{\theta}_u$ is sufficient if it retains the mutual information with x_u compared to the original observation $\mathbf{y}$.

A well-known sufficient observation for the detection of $\mathbf{x}$ is the received signal after MF [13]

$$\boldsymbol{\theta} = \mathbf{H}^{\mathrm{H}}\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{n}, \tag{14}$$

where $\mathbf{G} \triangleq \mathbf{H}^{\mathrm{H}}\mathbf{H} \in \mathbb{C}^{U\times U}$, and $\mathbf{n} \triangleq \mathbf{H}^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^U$. The (i,j)-th element of channel Gram matrix $\mathbf{G}$ is the inner product $g_{i,j} = \mathbf{h}_i^{\mathrm{H}}\mathbf{h}_j$. For $i \neq j$, the value of $|g_{i,j}|$ indicates the interference level between UT i and UT j [14]. If $\mathbf{h}_i$ and $\mathbf{h}_j$ are orthogonal, namely $g_{i,j} = 0$, then there is no interference between UT i and UT j. Building on BBCM, thanks to the beam domain channel sparsity, we have the following lemma. The proof is omitted due to the space limitation.

Lemma 1 *For $i, j \in \mathcal{Z}_U^+$, when $\mathcal{P}_i \cap \mathcal{P}_j = \varnothing$, we have*

$$\lim_{M\to\infty} g_{i,j} = 0. \tag{15}$$

Lemma 1 indicates that if the directional cosine sets of two UTs do not overlap, their channels tend to become orthogonal as the number of antennas M approaches infinity. Since the non-zero beam set is limited for each UT in HF skywave massive MIMO, the channels among most of the UT pairs exhibit asymptotic orthogonality, resulting in asymptotic sparsity of $\mathbf{G}$. Hereafter, we refer to such sparsity as *interference sparsity*, which is used for observation selection from the output of MF.

Next, we extract elements from $\boldsymbol{\theta}$ to construct an observation vector $\boldsymbol{\theta}_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\theta}$ for the detection of each symbol x_u, where $\mathbf{N}_u = [\mathbf{I}_U]_{:,\mathcal{N}_u} \in \mathbb{R}^{U\times N_u}$ and $\mathcal{N}_u \subseteq \mathcal{Z}_U^+$ are respectively UT selection matrix and UT set with $N_u = |\mathcal{N}_u|$. In order to construct a sufficient observation $\boldsymbol{\theta}_u$ for the detection of each transmitted symbol x_u, by making the assumption $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, we have the following theorem. The proof is omitted due to the space limitation.

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.

Theorem 1 *For a given observation vector $\boldsymbol{\theta}_u$,*

$$I(x_u, \boldsymbol{\theta}_u) \leq I(x_u, \mathbf{y}), \tag{16}$$

where the equality holds when $u \in \mathcal{N}_u$ and

$$g_{i,j} = 0, \ \forall i \in \mathcal{N}_u, j \notin \mathcal{N}_u. \tag{17}$$

As shown above, the orthogonality condition (17), the channels of UTs in a set containing UT u are orthogonal to those of UTs outside the set, can be asymptotically achieved when the number M of antennas is large, i.e., the interference sparsity. Thus, Theorem 1 basically says that with the interference sparsity, selecting the subvector of $\boldsymbol{\theta}$ corresponding to this UT set forms a sufficient observation for the detection of x_u.

According to the above discussion, we choose observation vector $\boldsymbol{\theta}_u$ for the detection of x_u with interference sparsity. To determine an effective metric of interference level between any pair of UTs, we denote channel correlation between UT i and j by normalizing $|g_{i,j}|$ as

$$\eta_{i,j} \triangleq \frac{\left|\mathbf{h}_i^{\mathrm{H}}\mathbf{h}_j\right|}{\|\mathbf{h}_i\|\|\mathbf{h}_j\|} = \frac{|g_{i,j}|}{\sqrt{g_{i,i}g_{j,j}}}. \tag{18}$$

Clearly, $0 \leq \eta_{i,j} \leq 1$, $\eta_{i,j} = \eta_{j,i}$, and $\eta_{i,j} = 0$ when $\mathbf{h}_i$ and $\mathbf{h}_j$ are orthogonal, $\eta_{i,j} = 1$ when $\mathbf{h}_i$ is a scaled version of $\mathbf{h}_j$. Therefore, channel correlation $\eta_{i,j}$ can serve as a metric of the interference level among UTs. A larger value of $\eta_{i,j}$ implies greater interference among UT i and UT j, and vice versa. Theorem 1 implies that the detection of x_u can be achieved without mutual information loss by only considering the interference from $\mathcal{N}_u$ if the condition (17) is satisfied. As mentioned earlier, with the interference sparsity, (17) approximately holds and thus we can construct a UT set with strong interference to UT u, and determine the observation for signal detection by the UT set. Specifically, we define the interference UT set (IUS) for UT u as

$$\mathcal{N}_u \triangleq \left\{u' \in \mathcal{Z}_U^+ \middle| \eta_{u,u'} > \eta_{\mathrm{th}}\right\} \text{ with } N_u = |\mathcal{N}_u|, \tag{19}$$

where $\eta_{\mathrm{th}} < 1$ is a preset threshold. Therefore, observation vector $\boldsymbol{\theta}_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\theta}$ can be expressed as

$$\boldsymbol{\theta}_u = \mathbf{G}_u\mathbf{x} + \mathbf{n}_u, \tag{20}$$

where $\mathbf{G}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{G}$, $\mathbf{n}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{n}$.

After determining observation vector $\boldsymbol{\theta}_u$, the interference sparsity-aware signal detection for UT u is expressed as

$$\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\boldsymbol{\theta}_u + q_u. \tag{21}$$

The MSE of the detection for UT u is given by

$$\widehat{\epsilon}_u \triangleq \mathbb{E}\left\{|x_u - \widehat{x}_u|^2\right\} = \mathbb{E}\left\{\left|x_u - \left(\mathbf{w}_u^{\mathrm{H}}\boldsymbol{\theta}_u + q_u\right)\right|^2\right\}. \tag{22}$$

By minimizing MSE, $\mathbf{w}_u$ and q_u can be obtained as [15]

$$\mathbf{w}_u = \left(\mathbf{G}_u\boldsymbol{\Sigma}\mathbf{G}_u^{\mathrm{H}} + \sigma_{\mathrm{z}}\mathbf{G}'_u\right)^{-1}\mathbf{G}_u\mathbf{e}_u\sigma_u, \tag{23a}$$

$$q_u = \mu_u - \mathbf{w}_u^{\mathrm{H}}\mathbf{G}_u\boldsymbol{\mu}, \tag{23b}$$

where $\mathbf{e}_u$ is the u-th column of $\mathbf{I}$, $\mathbf{G}'_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{G}\mathbf{N}_u$ and $\mathbf{w}_u$ is termed as the interference sparsity-aware detector (ISD) for UT u. Substituting (23) into (21) and (22) yields the signal reconstructed by ISD and the detection MSE for UT u:

$$\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\widetilde{\boldsymbol{\theta}}_u + \mu_u, \tag{24a}$$

$$\widehat{\epsilon}_u = \left(1 - \mathbf{w}_u^{\mathrm{H}}\mathbf{G}_u\mathbf{e}_u\right)\sigma_u, \tag{24b}$$

where $\widetilde{\boldsymbol{\theta}}_u = \mathbf{N}_u^{\mathrm{T}}\widetilde{\boldsymbol{\theta}}$ is a subvector of $\widetilde{\boldsymbol{\theta}} = \boldsymbol{\theta} - \overline{\boldsymbol{\theta}}$ with $\overline{\boldsymbol{\theta}} = \mathbf{G}\boldsymbol{\mu}$. Compared with the MMSE detection, the dimensionality of ISD $\mathbf{w}_u$ is much lower than MMSE detector $\mathbf{r}_u$, leading to lower computational complexity.

We next incorporate the ISD into the framework of turbo receiver, forming the interference sparsity-aware turbo receiver (ISTR). Using the ISD, the extrinsic information for SISO detection in turbo receiver can be obtained through (9) by replacing $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ with $\widehat{x}_u$ and $\widehat{\epsilon}_u$, respectively, yielding the extrinsic LLR $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$, which is conditioned on the low-dimensional observation $\boldsymbol{\theta}_u$ rather than the original observation $\mathbf{y}$. Other procedures are consistent with those discussed in III-A, then the ISTR is formulated.

C. Efficient Implementation of ISTR

In this subsection, we discuss the efficient implementation of ISTR and consider the case of even M, while the odd case follows a similar analytical approach. To begin with, we rewrite the beam matrix in (6) as

$$\mathbf{V} = -\frac{1}{\sqrt{M}}\mathbf{I}_{M,S}\mathbf{F}_S\boldsymbol{\Lambda}\overline{\boldsymbol{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A}, \tag{25}$$

where $\boldsymbol{\Lambda} = \mathrm{diag}\left\{\mathbf{I}_{S,2S}\mathbf{f}_{2S,M}^*\right\}$, $\overline{\boldsymbol{\Pi}}_S \triangleq [\mathbf{e}_2, \cdots, \mathbf{e}_S, -\mathbf{e}_1] \in \mathbb{R}^{S\times S}$. The structure of $\mathbf{V}$ in (25) is employed for efficient implementation of the ISTR.

Firstly, we discuss the low-complexity computation of ISD $\mathbf{w}_u$. According to the construction principle of IUS, the interference from UTs outside $\mathcal{N}_u$ to those within $\mathcal{N}_u$ is small. Consequently, we can approximate that $g_{i,j} \approx 0, \ i \in \mathcal{N}_u, j \notin \mathcal{N}_u$. Thus, observation $\boldsymbol{\theta}_u$ in (20) can be approximated by

$$\boldsymbol{\theta}_u = \mathbf{G}'_u\mathbf{x}_u + \sum_{u'\notin\mathcal{N}_u}\mathbf{G}_u\mathbf{e}_{u'}x_{u'} + \mathbf{n}_u \approx \mathbf{G}'_u\mathbf{x}_u + \mathbf{n}_u, \tag{26}$$

where $\mathbf{x}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{x}$ is the signal from IUS $\mathcal{N}_u$. In line with (23), $\mathbf{w}_u$ can be approximated as

$$\begin{aligned}\mathbf{w}_u &\approx \left(\mathbf{G}'_u\boldsymbol{\Sigma}'_u\mathbf{G}'_u + \sigma_{\mathrm{z}}\mathbf{G}'_u\right)^{-1}\mathbf{G}_u\mathbf{e}_u\sigma_u \\ &= \left(\boldsymbol{\Sigma}'_u\mathbf{G}'_u + \sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u\sigma_u,\end{aligned} \tag{27}$$

where $\boldsymbol{\Sigma}'_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{N}_u$. Moreover, we approximate the a priori variance $\boldsymbol{\Sigma}$ as $\sigma\mathbf{I}$ [12], [15], where $\sigma = \mathrm{tr}\{\boldsymbol{\Sigma}\}/U$ is the average variance. Then $\mathbf{w}_u$ can be further approximated as

$$\mathbf{w}_u \approx \left(\mathbf{G}'_u + q\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u, \tag{28}$$

where $q = \sigma_{\mathrm{z}}/\sigma$. Eq. (28) avoids a series of matrix multiplications, effectively reducing the computational complexity.

Next, we turn to the low-complexity computation of $\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\mathbf{N}_u^{\mathrm{T}}\widetilde{\boldsymbol{\theta}} + \mu_u$, where the computation of $\widetilde{\boldsymbol{\theta}} = \boldsymbol{\theta} - \overline{\boldsymbol{\theta}}$ can be divided into the computation of MF $\boldsymbol{\theta} = \mathbf{H}^{\mathrm{H}}\mathbf{y}$ and signal reconstruction $\overline{\boldsymbol{\theta}} = \mathbf{G}\boldsymbol{\mu}$, respectively. Using BBCM (7) and

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.

the structure (25) of the beam matrix, the process of MF can be expressed as

$$\boldsymbol{\theta} = \mathbf{D}^{\mathrm{H}}\mathbf{V}^{\mathrm{H}}\mathbf{y} = \frac{1}{\sqrt{M}}\mathbf{D}^{\mathrm{H}}\mathbf{I}_{A,S}\overline{\mathbf{\Pi}}_S^{(A-1)/2}\mathbf{\Lambda}^{*}\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{y}. \quad (29)$$

Observing (29), the computational complexity of MF mainly comes from matrix-vector multiplications by $\mathbf{F}_S$ and $\mathbf{D}$, where the computation of $\mathbf{F}_S^{\mathrm{H}}\mathbf{a}$ can be fast implemented by inverse fast Fourier transform (IFFT), and the computation of $\mathbf{D}^{\mathrm{H}}\mathbf{b}$ only involves a small number of scalar multiplications since beam domain channel matrix $\mathbf{D}$ exhibits significant sparsity. The a priori mean $\boldsymbol{\mu}$ is updated at each turbo iteration, then $\overline{\boldsymbol{\theta}}$ is updated accordingly, which can be approximately computed using the asymptotic sparsity of $\mathbf{G}$.

According to the above discussion, both the computation of the ISD and the computation of the signal reconstructed by ISD require channel Gram matrix $\mathbf{G} = \mathbf{H}^{\mathrm{H}}\mathbf{H}$. Then, we focus on the low-complexity computation of $\mathbf{G}$. Since accurate beam domain channel $\mathbf{D}$ is estimated for signal detection, then $\mathbf{G}$ can be computed by $\mathbf{G} = \mathbf{D}^{\mathrm{H}}\mathbf{Q}\mathbf{D}$ with $\mathbf{Q} \triangleq \mathbf{V}^{\mathrm{H}}\mathbf{V} \in \mathbb{R}^{A\times A}$. Utilizing the real-valued symmetric Toeplitz structure of $\mathbf{Q}$, the computation of $\mathbf{QD}$ can be derived as

$$\mathbf{QD} = \mathbf{D} + \mathbf{I}_{A,S}\sum_{n=1}^{\lceil S/2\rceil-1}\left(\overline{\mathbf{\Pi}}_S^{n} - \overline{\mathbf{\Pi}}_S^{S-n}\right)\mathbf{I}_{S,A}\mathbf{D}\gamma_n, \quad (30)$$

where $\gamma_n = \sin(\pi n/F)/(M\sin(\pi n/S))$. From (30), the computational complexity of $\mathbf{QD}$ arises from the computation of $\mathbf{D}\gamma_n$, only involving the multiplications of real-valued scalars γ_n and limited non-zero complex-valued elements in $\mathbf{D}$. As the number of antennas at the BS increases, S becomes huge, leading to a significant increase of terms of γ_n. To further reduce the computational complexity, a certain number of coefficients γ_n need to be filtered out. When $F > 1$, about $1/F$ of the terms of γ_n equal 0 and $\gamma_n \ll \gamma_0 = 1$ for most n. To this end, we define a filtering set

$$\mathcal{Q} \triangleq \left\{n \in \mathcal{Z}^{+}_{\lceil S/2\rceil-1} \middle| |\gamma_n| > \varepsilon\right\} \text{ with } Q = |\mathcal{Q}|, \quad (31)$$

where ε is a predefined threshold. Therefore, the computation of $\mathbf{QD}$ can be approximated by discarding the summation terms with $n \notin \mathcal{Q}$ in (30), i.e.,

$$\mathbf{QD} \approx \mathbf{P} = \mathbf{D} + \mathbf{I}_{A,S}\sum_{n\in\mathcal{Q}}\left(\overline{\mathbf{\Pi}}_S^{n} - \overline{\mathbf{\Pi}}_S^{S-n}\right)\mathbf{I}_{S,A}\mathbf{D}\gamma_n. \quad (32)$$

Once $\mathbf{P}$ is calculated, $\mathbf{G}$ can be approximately computed via $\mathbf{G} \approx \mathbf{D}^{\mathrm{H}}\mathbf{P}$ using its Hermitian structure and the sparsity of $\mathbf{D}$ and $\mathbf{P}$. Leveraging (32), it can be checked that $\left[\mathbf{D}^{\mathrm{H}}\mathbf{P}\right]_{i,j} = \left[\mathbf{D}^{\mathrm{H}}\mathbf{P}\right]_{j,i} = 0$ if $(i,j) \notin \mathcal{G}$, where the index set

$$\begin{aligned}\mathcal{G} = \Big\{(i,j) \in \mathcal{Z}_U^{+}\times\mathcal{Z}_U^{+} \Big| \, &(\mathcal{A}_i \cap \mathcal{A}_j) \cup (\mathcal{A}_i \cap \mathcal{A}_{j,n}) \\ &\cup (\mathcal{A}_i \cap \mathcal{A}_{j,-n}) \neq \varnothing, \ \forall n \in \mathcal{Q}\Big\}. \end{aligned} \quad (33)$$

In (33), non-zero beam set $\mathcal{A}_i = \{a_{i,1}, \cdots, a_{i,A_i}\}$, $\mathcal{A}_{i,n} = \{\langle a_{i,1}+n\rangle_S, \cdots, \langle a_{i,A_i}+n\rangle_S\}$, and $\langle\cdot\rangle_S$ is the modulo-S operation. Thus, the non-zero elements that need to be calculated in $\mathbf{G}$ can be pre-determined by $\mathcal{G}$ with number $G_{\mathrm{nz}} = |\mathcal{G}|$. The above efficient implementation of ISTR is concluded in Algorithm 1, where T is the number of iterations.

Algorithm 1: Efficient implementation of ISTR

Initialization: $t = 1$, $L'_{\mathrm{a}}(q_{u,i}) = 0, \ \forall u \in \mathcal{Z}_U^{+}, i \in \mathcal{Z}_N^{+}$.
Compute $\mathbf{G} = \mathbf{D}^{\mathrm{H}}\mathbf{P}$ with $\mathbf{P}$ computed by (32).
Compute $\boldsymbol{\theta}$ by (29).
while $t \le T$ **do**
  Compute $P(q_{u,i} = b)$, $\forall u \in \mathcal{Z}_U^{+}, \forall i \in \mathcal{Z}_N^{+}$, by (11) and update $\boldsymbol{\mu}$ and $\mathbf{\Sigma}$ by (12), respectively.
  Compute $\mathbf{w}_u, \ \forall u \in \mathcal{Z}_U^{+}$, by (28).
  Compute $\widehat{x}_u$, $\widehat{\epsilon}_u$, $\forall u \in \mathcal{Z}_U^{+}$, by (24).
  Compute $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$, $\forall u \in \mathcal{Z}_U^{+}, \forall i \in \mathcal{Z}_N^{+}$, by (9) with $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ replaced by $\widehat{x}_u$ and $\widehat{\epsilon}_u$, respectively.
  If $t < T$, update $L'_{\mathrm{a}}(q_{u,i})$, $\forall u \in \mathcal{Z}_U^{+}, \forall i \in \mathcal{Z}_N^{+}$, after deinterleaving, SISO decoding and interleaving.
  Set $t \leftarrow t + 1$.
end
Output $\widehat{c}_{u,i}$, $\forall u \in \mathcal{Z}_U^{+}, \forall i \in \mathcal{Z}_N^{+}$, after deinterleaving and SISO decoding.

IV. Simulation Results

In this section, simulation results of the proposed ISTR for HF skywave massive MIMO are provided.

UTs are distributed within a sector spanning 150° and located approximately $2,000$ km away from the BS and isotropic antennas are equipped at both ends. Some physical channel parameters, such as path gain and delay for each path from each UT, are generated by the ray-tracing software Proplab-Pro 3.1 [9]. The space domain channel at each sub-carrier is generated accordingly. The beam domain channel is obtained through the method proposed in [3] with fine factor $F = 2$. A row-column interleaver with length 528 is utilized for transmission. More detailed simulation parameters are presented in Table I.

TABLE I
Simulation Parameters

Parameter	Value
Carrier frequency f_{c}	16 MHz
Number of antennas at the BS M	256
Antenna spacing at the BS d	9 m
Number of UTs U	72
Modulation scheme	16-QAM
Channel code	3/4 LDPC code
Code length	2112

We adopt the bit-error rate (BER) as the performance metric for receivers. Fig. 1 compares the BER performance of the proposed ISTR with and without approximation, where the MMSE turbo receiver (MMSE TR) [15], the beam structured detector (BSD) [10], and the joint Slepian transform based detector (JSTD) [9] are included for comparison and the SNR represents the received signal-to-noise ratio. For ISTR, we set $\eta_{\mathrm{th}} = 0.03$ and $\varepsilon = 0.01$. Moreover, the ISTR with and

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.

without approximation refer to the ISTR using (32) and (28) for approximate computation and the ISTR using (30) and (23) for exact computation, respectively. From the figure, the BER performance of the ISTR with approximation approaches that of the ISTR without approximation, especially when the number of iterations $T = 3$, both approaching that of the MMSE TR. Therefore, unless otherwise specified, the ISTR hereinafter refers to the ISTR with approximation. At the BER of 10^{-3}, the ISTR with $T = 3$ achieves approximately a 2 dB gain compared with the MMSE TR with $T = 1$ and the JSTD, and more than 4 dB gain compared with the BSD.

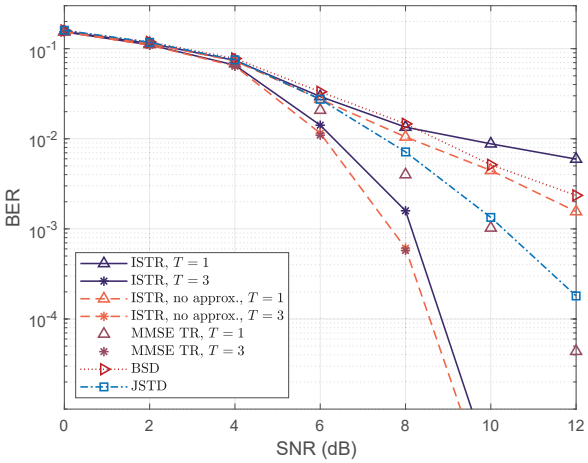


Fig. 1. BER performance of the ISTR with and without approximation, compared with the MMSE TR, BSD and JSTD.

In Fig. 2, the computational complexities of the ISTR and the other receivers are compared versus the number of UTs, where the number of complex multiplications is adopted as the metric, and $\varepsilon = 0.01$. For the ISTR without approximation, its complexity is below that of the MMSE TR and BSD and is lower than that of the JSTD in most cases, even at $T = 3$. Besides, compared with the ISTR without approximation, the ISTR with approximation exhibits a more pronounced complexity reduction and it is insensitive to the number of iterations.

V. Conclusions

In this paper, we investigated receiver design for HF skywave massive MIMO communications. We first introduced the BBCM and revealed the interference sparsity for HF skywave massive MIMO systems. Then, we discussed sufficient observation selection for signal detection by employing interference sparsity. As a result, the IUS was constructed and the ISTR was proposed with low-dimensional ISDs. Next, we proposed an efficient implementation of the ISTR by adopting the properties of the BBCM. Simulation results manifested the effectiveness of the proposed ISTR.

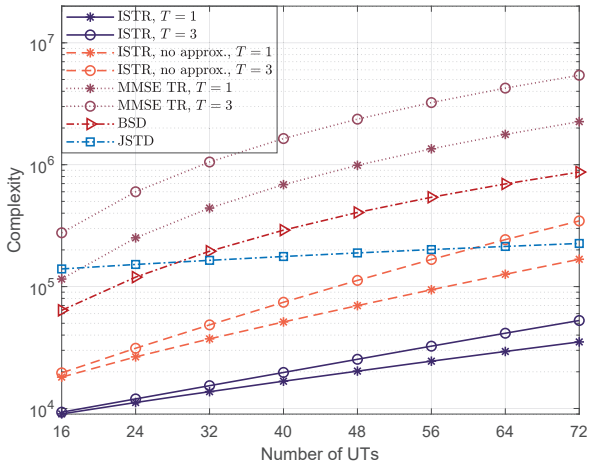


Fig. 2. Computational complexity of the ISTR, compared with the MMSE TR, BSD and JSTD.

References

[1] C. D. Alwis, A. Kalla, Q.-V. Pham, P. Kumar, K. Dev, W.-J. Hwang, and M. Liyanage, "Survey on 6G frontiers: Trends, applications, requirements, technologies and future research," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 836–886, Apr. 2021.

[2] X. L. Yu, A.-A. Lu, X. Q. Gao, G. Y. Li, G. Ding, and C.-X. Wang, "HF skywave massive MIMO communication," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2769–2785, Oct. 2021.

[3] D. Shi, L. Song, W. Zhou, X. Q. Gao, C.-X. Wang, and G. Y. Li, "Channel acquisition for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4074–4089, Jun. 2023.

[4] L. Song, D. Shi, L. Gan, and X. Q. Gao, "Robust precoding for HF skywave massive MIMO with Slepian transform," *IEEE Trans. Commun.*, vol. 72, no. 6, pp. 3752–3765, Jun. 2024.

[5] M. A. Albreem, M. Juntti, and S. Shahabuddin, "Massive MIMO detection techniques: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3109–3132, Aug. 2019.

[6] M. Tuchler, R. Koetter, and A. Singer, "Turbo equalization: Principles and new results," *IEEE Trans. Commun.*, vol. 50, no. 5, pp. 754–767, May 2002.

[7] M. Abdelghany, U. Madhow, and A. Tölli, "Beamspace local LMMSE: An efficient digital backend for mmwave massive MIMO," in *Proc. IEEE 20th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Jul. 2019, pp. 1–5.

[8] T. Takahashi, A. Tölli, S. Ibi, and S. Sampei, "Low-complexity large MIMO detection via layered belief propagation in beam domain," *IEEE Trans. Wireless Commun.*, vol. 21, no. 1, pp. 234–249, Jan. 2022.

[9] L. Song, D. Shi, L. Gan, and X. Q. Gao, "Slepian transform based detectors for HF skywave massive MIMO-OFDM systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 13 106–13 119, Oct. 2023.

[10] D. Shi, L. Song, X. Q. Gao, J. Wang, M. Bengtsson, and G. Y. Li, "Beam structured signal detector for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 19 459–19 474, Dec. 2024.

[11] Z. Qin, J. Wang, J. Chen, G. Ding, Y.-D. Yao, X. Ji, and X. Chen, "Link quality analysis based channel selection in high-frequency asynchronous automatic link establishment: A matrix completion approach," *IEEE Syst. J.*, vol. 12, no. 2, pp. 1957–1968, Jun. 2018.

[12] Q. Guo and D. D. Huang, "A concise representation for the soft-in soft-out LMMSE detector," *IEEE Commun. Lett.*, vol. 15, no. 5, pp. 566–568, May 2011.

[13] B. Borloz and B. Xerri, "Subspace SNR maximization: The constrained stochastic matched filter," *IEEE Trans. Signal Process.*, vol. 59, no. 4, pp. 1346–1355, Apr. 2011.

[14] T. Weber, A. Ahrens, and S. Deng, "Decentralized interference cancellation in mobile radio networks," in *2007 IEEE Wireless Communications and Networking Conference*, Mar. 2007, pp. 2190–2194.

[15] M. Tuchler, A. Singer, and R. Koetter, "Minimum mean squared error equalization using a priori information," *IEEE Trans. Signal Process.*, vol. 50, no. 3, pp. 673–683, Mar. 2002.

Authorized licensed use limited to: Southeast University. Downloaded on July 21,2026 at 06:06:19 UTC from IEEE Xplore. Restrictions apply.